



Corneal Tomography Multivariate Index (CTMVI) effectively distinguishes healthy corneas from those susceptible to ectasia

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ABSTRACT

Background: Keratoconus (KC) is a multifactorial progressive corneal disease, pathogenically characterized by genetic, biochemical, and environmental key factors. Nevertheless, current tools used for its diagnosis, such as Pentacam, examine the posterior and anterior surfaces of the cornea based on a rotating Scheimpflug camera and provide, as output, a set of features for which the clinical interpretation has been error-prone.

Objective: Consequently, the objective of this study is to describe a novel Artificial Intelligence (AI) strategy that receives subsets of the 52 original Pentacam tomographic descriptors as input and produces, as output, the scalar value called *Corneal Tomography Multivariate Index* (CTMVI). Specifically based on Paraconsistent Feature Engineering (PFE) and Support Vector Machine (SVM), it intends to help clinicians in the pre-diagnosis of cornea ectasia risk assessment.

Methods: We collected data from patients' eyes in three different classes: 411 with healthy corneas to compose the control group (CG), 302 with KC to represent the KC group (KCG) and, additionally, 64 in a group with very asymmetric ectasia but with normal corneal topography (VAE-NTG) in one eye, totaling 777 samples. A set of features from the patients' corneal tomographic assessments was extracted to serve as input to the paraconsistent feature selector algorithm. Once it provides the best subsets of features representing the classes, a one-against-all SVM-based algorithm was successfully used to classify those groups. Notably, normal topography criteria (NTPC) were considered based on the steepest front curvature (K_{max}) < 47.2 diopters, a paracentral inferior-superior (I-S value) < 1.45, and a keratoconus percentage index (KISA%) score < 60. Normal corneal tomography (NTMC) consists of an anterior chamber depth < 3.8 μm, a front corneal apical elevation < 4 μm, a front corneal elevation (FCE) at the thinnest point < 3.8 μm, and an FCE < 12 μm in the corneal 4.0 mm center. Respective back corneal elevation values were 7 μm, 13 μm, and 25 μm. To assess the clinical validity of the models and their abilities to correctly classify new data, two procedures were used: the first was a holdout cross validation, and the second was the external validation consisting of an independent test with samples that were not part of the training data.

Results: Notably, PFE-based analysis effectively found dominant representative features which allowed for relevant classification accuracies based on SVM. Among numerous results, a parallel test combining CTMVI and existing Pentacam Random Forest Index (PRFI) produced a value of sensitivity of 0.99 and a value of specificity of 0.84, using 50% of the dataset for system modeling and the remaining 50% for assessment, improving current reports found in the literature. CTMVI, BAD-D, and PRFI showed an accuracy (acc) of 0.89, 0.86, and 0.89, respectively, in the external validation population to distinguish the healthy eyes from subclinical cases. The three indexes resulted in 0.99 acc when used to distinguish healthy eyes from corneal ectasia ones. CTMVI showed 100% sensitivity and 100% specificity between CG and KCG (cutoff ≤ -1.36). Regarding VAE-NTG patients, CTMVI had an area under the curve (AUC) of 0.935 (87.5% sensitivity; 84.95% specificity; cutoff ≤ 0.639), which was higher than those of Belin/Ambrósio Enhanced Ectasia Total Deviation Value (BAD D) with AUC 0.765; 78.12% sensitivity; 65.05% specificity; $p = .0001$, and PRFI with AUC 0.805;

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59.38% sensitivity; 90.78% specificity; $p = .0004$. During the external validation, it was shown to be as effective as BAD and PRFI.

Conclusions: Based on the innovative joint application of PFE and SVM to distinguish healthy corneas from those pathologically-affected, we provided an original contribution to the state-of-the-art in terms of AI-aided diagnosis. Therefore, CTMVI improves the detection of corneal ectasia susceptibility using tomographic-derived descriptors. This reassures the efficacy of the proposed approach.

1. Introduction

Corneal Ectasia (CE) is characterized by a progressive distortion of corneal curvature, which is thought to be associated with a “weak” cornea. Conceptual models of keratoconus pathogenesis have evolved considerably in recent years and have postulated keratocyte apoptosis [1]. Keratoconus (KC) is a multifactorial progressive corneal disease [2,3], where genetic, biochemical and environmental factors play a key role in its pathogenesis [4]. Diagnosing such an ectatic disorder is challenging in its early stages when the cornea looks healthy, even with the use of modern screening devices. This is because protrusion and thinning can only be seen with slit-lamp examinations in their later stages [5].

Post-refractive surgery ectasia, first described by Seiler and Quurke in 1998, has been recognized as a rare complication capable of further deteriorating vision after Laser-Assisted In-situ Keratomileusis (LASIK) [6,7]. Thus, corneal ectasia is, indeed, one of the most serious complications of this surgery. The risk factors already identified in post-LASIK ectasia include high thin residual stromal bed, high myopia, defined topographic abnormalities, genetic predisposition to KC, eye rubbing, young age, and pregnancy [8–10]. The percent tissue altered (PTA), a newly proposed metric has been reported to be a robust indicator for ectasia risk calculation, where $PTA > 40\%$ [11–13]. However post-refractive surgery ectasia can also occur after Photorefractive Keratectomy (PRK) and Small Incision Lenticule Extraction (SMILE) [14, 15]. Its prevention has, each time more, become a major concern for refractive surgeons [16]. Ectasia occurs owing to biomechanical decompensation of the corneal stroma, perhaps associated with severe damage to the corneal structure, i.e., high myopia treatment, or preoperative changes in its biomechanical properties. The current comprehension is that a combination of those factors possibly determines the stability or progression of ectasia after Laser Vision Correction (LVC) [16].

Novel imaging technologies, such as those based on biomechanical assessment and tomography, have proven to considerably enhance the ability to detect mild ectatic diseases, including those occurring in eyes exhibiting normal topography from subjects with clinical ectasia in the fellow eye [16]. Scheimpflug tomography, which reveals three-dimensional maps, posterior and anterior surfaces of the cornea, pachymetry and characteristics of the anterior chamber, is one example technique. Particularly interesting is Pentacam Scheimpflug technology, which involves the application of a rotating Scheimpflug camera to examine 138000 distinct elevation points of the posterior and anterior surfaces in the cornea [17]. The Belin/Ambrósio Enhanced Ectasia Total Deviation Value (BAD-D), which is derived from Pentacam and combines standard and enhanced elevation maps [16,18–20] of the back and front surfaces with thickness distribution data [20], is another useful tool that was designed to screen patients prior to refractive surgery. Nevertheless, subclinical aspects of KC are still obscure in terms of an accurate diagnosis, stimulating further research [5].

As observed in paper [21], machine learning and artificial intelligence (AI) algorithms play a key role in further improving ectasia risk assessment, helping us to deal with large amounts of information. The initial applications of AI to diagnose ectasia were published in 1994 and 1995 [22,23]. Specifically, Pentacam Random Forest Index (PRFI) is an AI-based parameter that identifies ectasia and its susceptibility, with relevant accuracy to detect KC [21]. Moreover, PRFI has shown a significant role in the detection of post-LASIK ectasia and highly susceptible corneas to ectasia (HSCE) [21].

To further evolve current AI-related strategies, such as those mentioned, Paraconsistent Feature Engineering (PFE) [24] is innovatively used in this paper to select the best subset of features from real patients’ data, aiming at the distinction of three classes: healthy corneas, composing the control group (CG), corneas with KC, representing the KC group (KCG) and, complementarily, the group of corneas presenting very asymmetric ectasia but with normal corneal topography (VAE-NTG) in one eye. Then, those representative features served as input to a Support Vector Machine (SVM) [24–26], successfully classifying with relevant sensibilities and specificities, the proportion of 50% of our database: a total of 777 patients, i.e., 411 from CG, 302 from KCG and 64 from VAE-NTG.

Focusing on a careful exposition of the concepts and ideas exposed hereafter, this article is organized as follows. Section 2 fully describes the proposed approach, Section 3 reports the corresponding tests and results, Section 4 presents the discussion where our research results are contrasted with the state-of-the-art and, lastly, the conclusions, that are followed by the references, are presented in Section 5. This piece of research is illustrated with relevant figures and is enriched with informative tables in order to provide clear and complementary explanations.

2. Materials and methods

2.1. The dataset

The criteria for clinical diagnosis of healthy and pathologically-affected eyes included topographic characteristics, such as inferior steepening or skewed asymmetric bow-tie, and, at least, one slit-lamp finding, such as Fleischer’s ring, Munson’s sign, Vogt’s striae, Rizutti’s sign or an apical thinning. Patients were carefully considered to be very asymmetric (VAE-NTG) whenever the diagnosis of ectasia was confirmed in one eye, based on the previously described criteria, and the fellow eye had a normal front surface curvature, i.e., topometric map. Objective criteria we adopted for considering normal topography were applied to define VAE-NTG, including an objective front surface curvature set of metrics derived from Pentacam. Normal topography was rigorously considered based on the objective criteria [21,27] of a maximum curvature, i.e., Steepest Front Keratometry (Kmax), lower than 47.2 diopters, a paracentral inferior–superior (I–S value) asymmetry value at 6 mm, i.e., 3-mm radii, lower than 1.45, and keratoconus percentage index (KISA score) lower than 60 [28]. Normal corneal tomography (NTMC) for CG and VAE-NT were the maximum values of 3.8 mm for anterior chamber depth, 4 μm for front apical elevation, 5 μm for front corneal elevation (FCE) at the thinnest point and 12 μm for FCE in the central 4.0 μm . In addition, the corresponding posterior elevation values were 7 μm , 13 μm and 25 μm [29].

As mentioned above, our dataset includes data from a total of 777 patients distributed as follows:

- **CG:** these are the normal, or healthy, eyes. This group comprised data from eyes of 411 patients who underwent LASIK or PRK, stable after at least 18 months of follow-up, without any changes in the posterior elevation at the 18-month Pentacam assessment in relation to the preoperative test. The corresponding surgeries were performed from 2012 to 2018. The inclusion criteria for normality were:
 - healthy corneas observed during eye inspection, in both eyes, with a normal slit-lamp biomicroscopy;

- a value of corrected distance visual acuity of 20/20, or better;
- an overall normal tomography and topography inspections.

The 411 patients from this group were considered with healthy corneas with NTPC and NTMC in both eyes. Just one eye was selected in a random manner for further statistical analysis. Particularly, one eye per patient was randomly included to avoid selection bias which could be related to the selection of both eyes from the same patient.

- **KCG:** this group comprised 302 patients, where one eye from each one was considered, with bilateral clinical KC. As in the previous case, just one eye was included per patient, in a random manner, to avoid the selection bias related to the incorrect use of both eyes from the same patient. The inclusion criteria were eyes of KC displaying a KISA% index higher than 100% and, at least, one of the following biomicroscopic signs: Fleischer ring, focal stromal thinning or Vogt striae. Oculus implemented their specific staging system into the Pentacam control software, which is expected to mimic the Amsler/Krumeich systems, i.e., the Topographic Keratoconus Classification (TKC) [30]. TKC classifies KC into four stages, plus four intermediate stages. The TKC classification of Pentacam tomographic index shows its degree of KC ranging from 1 to 4.
- **VAE-NTG:** as mentioned above, this group comprised data from 64 eyes, from 64 different patients, with very asymmetric ectasia and with NTPC and NTMC in one eye and frank ectasia in the fellow eye. The eye that was included had a feature of CG. The inclusion criteria were based on previous studies documented in papers [21,31], and [32]. Eyes with insufficient topographic findings to meet the diagnostic criteria for keratoconus and features such as normal-appearing cornea on slit-lamp biomicroscopy, retinoscopy and keratometry were also included in this group. Specifically, we have in this group:
 - less affected fellow eye of a keratoconic patient where KISA% index is lower than 60%, the I-S difference value is lower than 1.45 D and, in addition, Kmax value is lower than 47.2 D, i.e., just the same topographic criteria we adopted for healthy eyes, but observing that, in normal organs, both patients' eyes met the required criteria. These patients can be considered to have HSCE.
 - fellow eyes of VAE-NT cases with the same criteria of KCG.

All the subjects underwent complete eye examination as well as refraction, biomicroscopy, retinoscopy, funduscopy, topography and tomography assessments. Accordingly, all patients were assessed at Visum Eye Center¹ from January 2012 to January 2018. The following exclusion criteria were adopted: history of ocular trauma; chronic use of eye medication and glaucoma; dry eye syndrome; corneal scarring; neurotrophic keratopathy; severe meibomian gland dysfunction; vulnerable state owing to physical or mental illness and with language-related difficulties; pregnancy or breastfeeding. According to the standards of the participating clinics, contact lens wearing was discontinued at least three weeks before the assessment for required rigid contact lenses and one week for required soft contact lenses.

Particularly, this dataset and its associated experiments adhered to the tenets of the Declaration of Helsinki and were approved by the Research Ethics Committee of São José do Rio Preto Faculty of Medicine (CAAE/FAMERP; process no. 1,123,652) and protocol registration on U.S. National Library of Medicine clinical trials (process no. NCT04313387). All patients were informed about the objectives of our study and they signed written informed consent forms before being enrolled.

All eyes were examined based on Pentacam Tomography by the rotation of Scheimpflug corneal and anterior segments, i.e., Pentacam HR from Oculus GmbH, Wetzlar, Germany. Images were checked in such a way that only cases with a rigid acceptable quality were included in our study. An experienced and well-known fellowship-trained corneal specialist (GCAJ) reviewed all the cases in such a way that they were correctly classified in as belonging to the KC and VAE-NTG classes. For inclusion in our study, all paired eyes necessarily have at least 9.0 mm of coverage considering a minimum of 8.0 mm nonextrapolated data points and an intrinsic Pentacam quality inspection indicating no issues. The raw data, i.e., u12 files, were obtained from all cases. Therefore, the same customized software, for which version is 1.20r118, was used to process all the files. Accordingly, all Scheimpflug variables were directly extracted from the Pentacam software using the “call-all” function.

Complementarily and just to illustrate samples of our database, Figs. 1 and 2 show corneal tomographies from patients with an eye possessing subclinical ectasia and fellow eye possessing frank ectasia. The aspect of the left eyes of these two patients is similar to one of a healthy cornea. Notice that the aspect of color grading on axial curvature front on the left eyes is more homogeneous than the one of the right ones.

2.2. The proposed AI-based strategy

The proposed system architecture, shown in Fig. 3, is composed by the following modules: Pentacam assessment, PFE-based feature selection, SVM classifier and Corneal Tomography Multivariate Index (CTMVI) system output. Pentacam provides, for each eye under analysis, the following intrinsic features ($F_0, F_1, \dots, F_{52}$) as a result of its assessment, where all Zernike measurements were made for a corneal diameter of 6 mm:

- F_0 : anterior radius of curvature (ARC) in the 3.0-mm zone centered on the thinnest location of the cornea;
- F_1 : Ambrósio relational thickness maximum (ARTmax);
- F_2 : Ambrósio relational thickness minimum (ARTmin);
- F_3 : Belin/Ambrósio enhanced ectasia total deviation value (BAD-D);
- F_4 : deviation of the ART average (BAD-Daa);
- F_5 : deviation of the ART max (BAD-Dam);
- F_6 : deviation of back elevation difference map (BAD-Db);
- F_7 : deviation from the posterior elevation at the thinnest considering BFS 8 mm (BAD-De);
- F_8 : deviation of front elevation difference map (BAD-Df);
- F_9 : deviation of minimum thickness (BAD-Dt);
- F_{10} : deviation from Kmax (BAD-Dk);
- F_{11} : deviation of average pachymetric progression (BAD-Dp);
- F_{12} : deviation from the more negative value on the relative thickness map (BAD-Dr);
- F_{13} : deviation from the vertical displacement of the thinnest point from the apex (BAD-Dy);
- F_{14} : corneal volume of 3-mm diameter area (C.Vol D 3 mm);
- F_{15} : corneal volume of 5-mm diameter area (C.Vol D 5 mm);
- F_{16} : corneal volume of 7-mm diameter area (C.Vol D 7 mm);
- F_{17} : corneal volume of 10-mm diameter area (C.Vol D 10 mm);
- F_{18} : $\frac{D2 \text{ mm}}{\text{Pachy Min}}$, where the numerator and denominator are defined ahead;
- F_{19} : average corneal thickness of 2 mm circle centered on the thinnest location (D2 mm);
- F_{20} : $\frac{D4 \text{ mm}}{\text{Pachy Min}}$, where the numerator and denominator are defined ahead;
- F_{21} : average corneal thickness of 4-mm circle centered on the thinnest location (D4 mm);
- F_{22} : $\frac{D6 \text{ mm}}{\text{Pachy Min}}$, where the numerator and denominator are defined ahead;

¹ <http://visum.med.br>

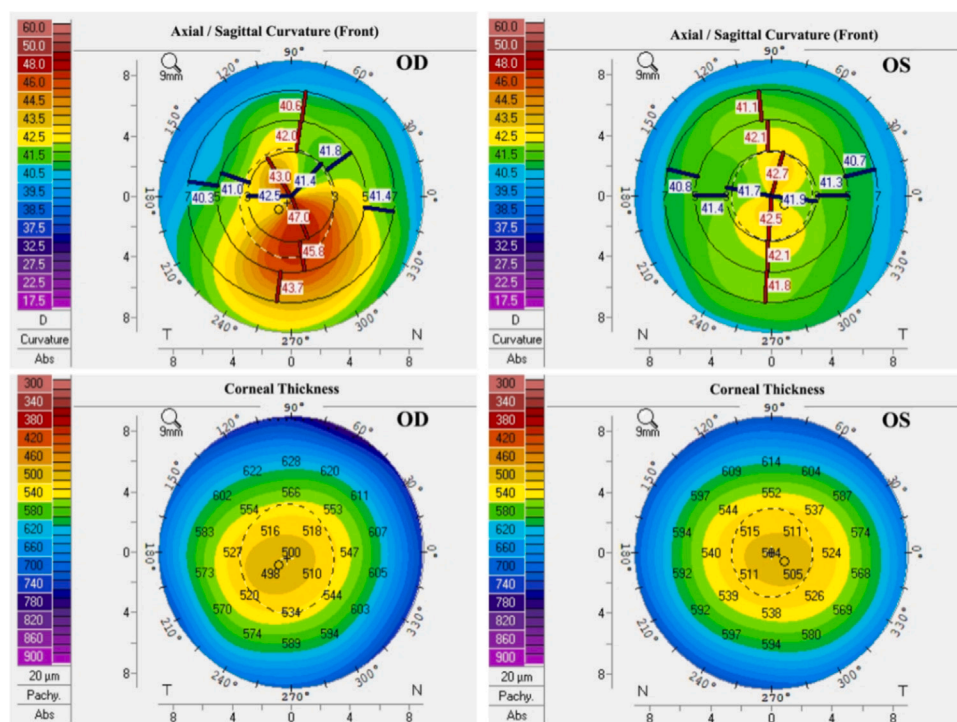


Fig. 1. Corneal tomography of a patient with very asymmetric keratoconus showing (OD) ectasia and (OS) regular axial curvature (front) on tomography in the fellow eye. Corneal thickness of both the eyes is similar.

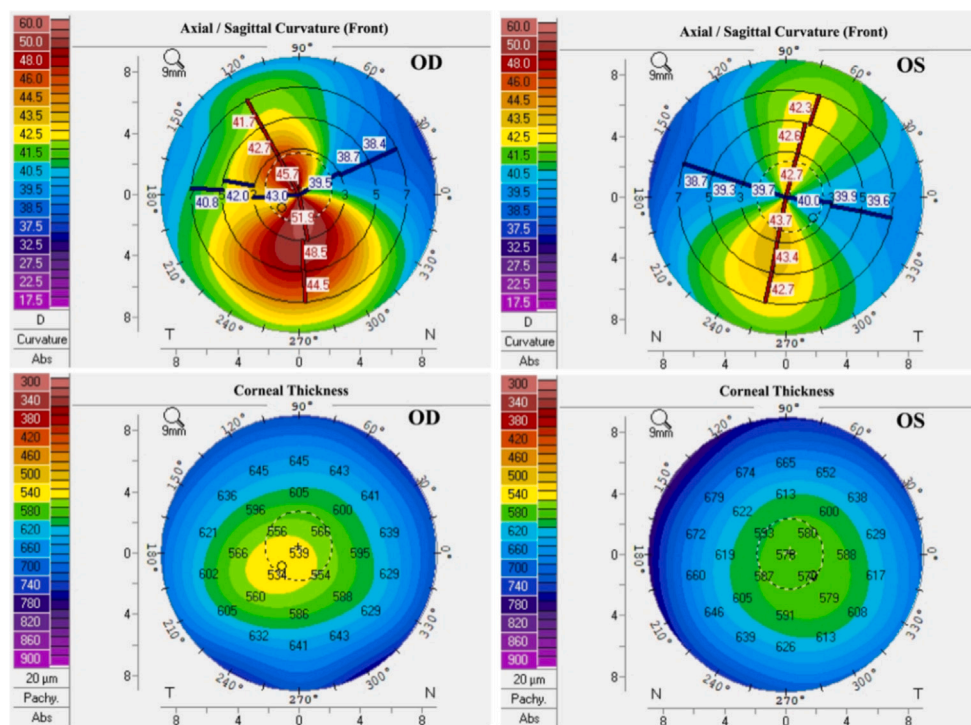


Fig. 2. Corneal tomography of patient with very asymmetric keratoconus showing (OD) ectasia and (OS) regular axial curvature (front) on tomography in the fellow eye. The thinnest points from both the eyes are different.

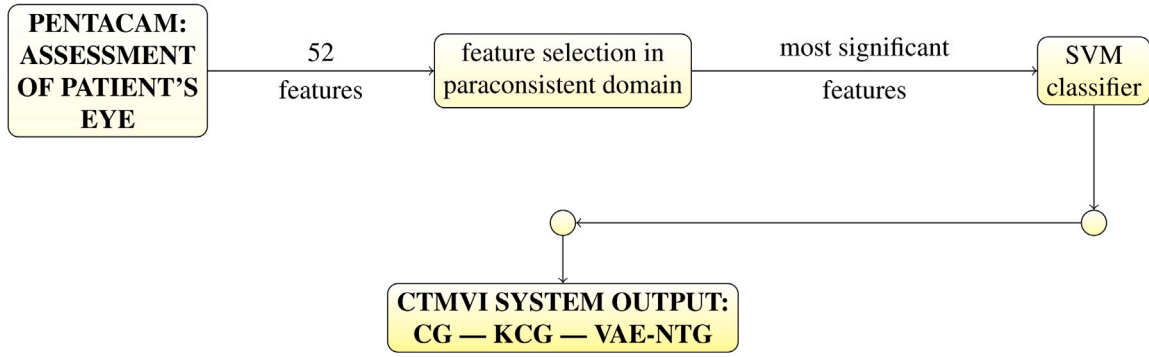


Fig. 3. The proposed approach structure.

- F_{23} : average corneal thickness of 6 mm circle centered on the thinnest location (D6 mm);
- F_{24} : $\frac{D8 \text{ mm}}{\text{Pachy Min}}$, where the numerator and denominator are defined ahead;
- F_{25} : average corneal thickness of 8 mm circle centered on the thinnest location (D8 mm);
- F_{26} : elevation parameter derived from the back surface centered at the point with highest value within the 4-mm diameter using the 8-mm best-fit toric ellipsoid (Ele B BFTE 8 mm Max. 4 mm Zone);
- F_{27} : elevation parameter derived from the back surface centered at the apex calculated using the 8-mm best-fit toric ellipsoid (Ele B BFTE 8 mm Apex);
- F_{28} : elevation parameter derived from the back surface centered at the thinnest point using the 8-mm best-fit toric ellipsoid (Ele B BFTE 8 mm Thinnest);
- F_{29} : elevation parameter derived from the front surface centered at the point with highest value within the 4-mm diameter using the 8-mm best-fit toric ellipsoid (Ele F BFTE 8 mm Max. 4 mm Zone);
- F_{30} : elevation parameter derived from the front surface centered at the apex calculated using the 8-mm best-fit toric ellipsoid (Ele F BFTE 8 mm Apex);
- F_{31} : elevation parameter derived from the front surface centered at the thinnest point using the 8-mm best-fit toric ellipsoid (Ele F BFTE 8 mm Thinnest);
- F_{32} : index highest asymmetry (IHA);
- F_{33} : index highest decentration (IHD);
- F_{34} : index of surface variance (ISV);
- F_{35} : index of vertical asymmetry (IVA);
- F_{36} : keratoconus index (KI);
- F_{37} : corneal thickness at the thinnest point (Pachy Min);
- F_{38} : position of minimum corneal thickness in relation of Y axis centered on cornea apex (Pachy Min Y);
- F_{39} : posterior radius of curvature in the 3.0-mm zone centered on the thinnest location of the cornea (PRC 3 mm Zone);
- F_{40} : relative corneal thickness at the thinnest point (Rel Pachy Min);
- F_{41} : root mean square of high order aberration of cornea back (RMS HOA CB);
- F_{42} : root mean square of high order aberration of cornea front (RMS HOA CF);
- F_{43} : root mean square of high order aberration of total cornea (RMS HOA Cornea);
- F_{44} : average pachymetric progression index (RPIavg);
- F_{45} : maximum pachymetric progression index (RPImax);
- F_{46} : minimum pachymetric progression index (RPImin);
- F_{47} : 3rd order vertical coma aberration cornea back (Z 3 -1 CB);
- F_{48} : 3rd order vertical coma aberration of cornea front (Z 3 -1 CF);

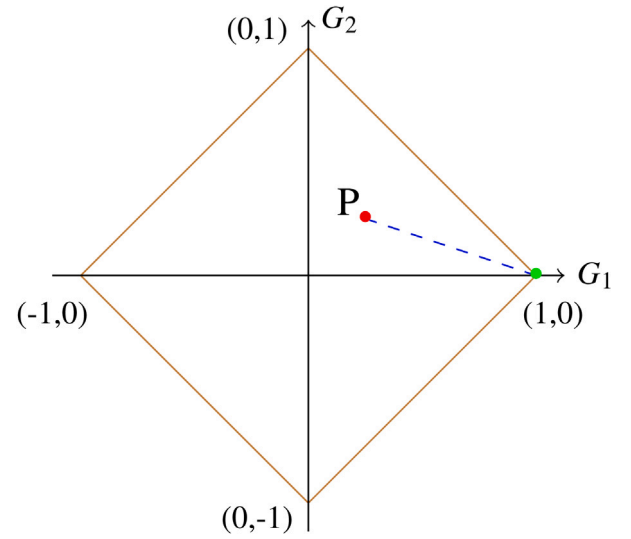


Fig. 4. The paraconsistent plane, where the axes G_1 and G_2 characterize the degrees of certainty and contradiction, respectively. The point $P = (G_1, G_2) = (\alpha - \beta, \alpha + \beta - 1)$, arbitrarily drawn in red just as an example, is mandatory for the analyses: the closer it is to the corner (1,0), drawn in green, the better the feature vectors separate between the classes, disregarding the classifier associated with them. The blue dashed line represents the distance from P to (1,0).

- F_{49} : 3rd order vertical coma aberration total cornea (Z 3 -1 Cornea);
- F_{50} : 5th order vertical coma aberration of cornea back (Z 5 -1 CB);
- F_{51} : 5th order vertical coma aberration of cornea front (Z 5 -1 CF).

In our experiments, Pentacam produced one feature vector of dimension 52 for each eye examined, totaling 777 vectors, i.e., $F_1[\cdot], F_2[\cdot], \dots, F_{777}[\cdot]$. They served as input to our PFE-based approach, used to select the most significant subsets of features. Essentially, the procedure starts with the analysis of two independent indexes, as detailed in [26], considering all the vectors of the three classes together: α and β , describing the lowest degree of intraclass similarity and the degree of interclass dissimilarity, respectively, where $(0 \leq \alpha, \beta \leq 1)$. Then, $G_1 = \alpha - \beta$ and $G_2 = \alpha + \beta - 1$, reflecting the degree of certainty and the degree of contradiction, respectively, must be obtained. Lastly, the point $P = (G_1, G_2)$ is placed in the paraconsistent plane, shown in Fig. 4, and its distance from the optimum corner (1,0), which means maximum intraclass similarity and no interclass overlap, is calculated.

In addition to the complete feature vectors with 52 elements each one, **all** the possible subsets with 1, 2, $\dots$, 51 features were assessed, always considering the entire database with 777 vectors. The best subsets

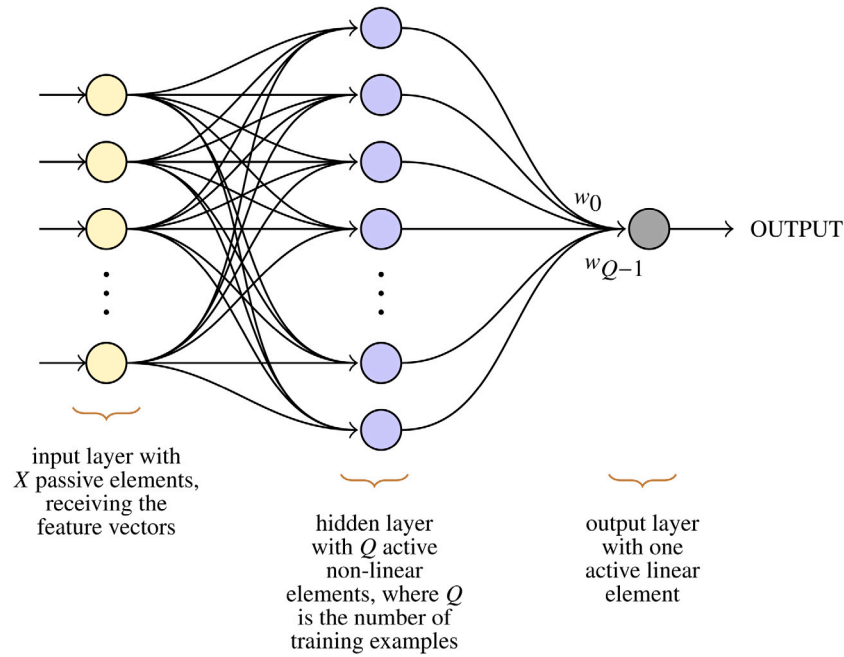


Fig. 5. SVM structure used in the proposed approach, considering the one-against-all strategy. The set of weights, i.e., $\{w_0, w_1, \dots, w_{Q-1}\}$ is determined during the supervised step of the training procedure.

of features correspond to those placing P closer to (1,0). Since PFE-related procedures are very well described in paper [26], no further explanation about them have been included here.

Once the most significant subsets of features were obtained, a one-against-all SVM structure [25,26], as shown in Fig. 5, was adopted to classify the feature vectors isolated for testing the proposed approach in just one of the three classes: CG, KCG or VAE-NTG. The SVM was defined with three layers: the input layer with X passive elements, the hidden layer with Q active non-linear elements and, lastly, the output layer with one active linear element, where $(0 < Q \leq 52)$ is the dimension of the feature vectors and X corresponds to the number of training vectors. The value of X was chosen in that way in order to produce a linear system of X equations in X unknowns, directly solvable. The hidden elements use Radial Basis Functions (RBFs) and were trained based on a non-supervised procedure where means and standard-deviations were set based on training data. Contrary to this, the output element simply balances the results from the hidden elements, where the corresponding weights $(w_0, w_1, \dots, w_{Q-1})$ are determined based on a supervised procedure using the training data. Each output neuron was set to either 1 or -1 during supervised step to indicate the class that the input vector belongs to.

The classifier output reflects the final results for the input vector. Particularly, the corresponding main statistical metrics were sensitivity and specificity, considering a hold-out procedure, as explained in the next section. Python language [33] was used in all data manipulation. Particularly, NumPy and Scikit-learn packages were used to test the proposed learning models.

2.3. Complementary information

In addition to the above-described procedures, it is important to register that statistical distributions were assessed carefully by using the Kolmogorov–Smirnov goodness-of-fit and Anderson–Darling. Almost all data in the three groups were, originally, not normally distributed. Therefore, comparisons between them were carried out with the nonparametric Kruskal–Wallis test, subsequently followed by post-hoc Dunn’s test to contrast each pair of groups. Receiver Operator Characteristic (ROC) curves were analyzed to determine the sensitivities and specificities, where the corresponding Areas Under the Curves

(AUCs) were used as measures of accuracy. DeLong’s [34] test was performed to pairwise compare different AUCs and the binomial exact test was used to observe the 95% confidence intervals (CI). ROC curves were calculated based on 95% CI and the results were reported as percentages of specificity and sensitivity. Significance was set at 0.05. ROC curves were calculated to assess the ability of CTMVI to differentiate among patients in the classes. Fisher’s exact test was adopted to compare sensitivity and specificity involving CTMVI, BAD-D and PRFI in the external validation results. Sensitivity, specificity, accuracy and cut-off values of CTMVI, BAD D, and PRFI were calculated from ROC curves with Youden’s index. The confidence interval of Youden index, and cut-off value were evaluated by bootstrap method (1000 iterations; random number seed: 978). GraphPad Prism 8 version 8.1.2² and MedCalc³ were used for statistical analysis and graph building.

3. Statistical analyses, tests and results

3.1. General statistics over the database

We register that the mean age of patients is 34.44 ± 10.0 years for CG, 28.42 ± 8.68 years for VAE-NTG and 29.51 ± 11.17 years for KCG. The ages differed significantly among the groups, with a p -value lower than 0.0001, where GC has the highest value. As assessed by pairwise post-hoc Dunn’s test comparisons, just VAE-NTG vs KC have not differed significantly. The male-to-female ratios differed significantly among the three groups, with a p -value = 0.001. Table 1 summarizes that information. Additionally, we can interestingly note that the three groups showed statistically-significant differences for all the features, as shown in Table 2.

3.2. The best subsets of features

After obtaining general statistics over the database, as complementary information only, PFE was used to determine the best subsets of features, from the 52-sample long features vectors, to distinguish

² Graph Pad Software, San Diego, USA; <https://www.graphpad.com>; 2019.

³ MedCalc Software bvba, Ostend, Belgium; <http://www.medcalc.org>; 2017.

Table 1

Age, gender, and sample number of the patients included in the study.

	CG	VAE-NTG	KCG	Remarks
Age	34.35 ± 9.88	29.94 ± 9.53	30.51 ± 11.96	* $P < 0.0001$ **CG · VAE-NTG ($P = 0.004$) **CG · KCG ($P < 0.0001$) **VAE-NTG · KCG (NS)
Number of patients	411	64	232	
Male patients	134	28	115	
Female patients	277	36	117	***CG · VAE-NTG (NS) ***CG · KCG ($P < 0.0001$) ***VAE-NTG · KCG (NS)

*Kruskal–Wallis test, ** Dunn's multiple comparisons test, ***Fisher's exact test CG: control group; VAE-NTG: very asymmetric ectasia with normal corneal topography + VAE with ectasia group; KCG: keratoconus group; NS: not significant.

Table 2

Distribution of each feature, where “SD” means “standard deviation”, “Min” means “minimum”, “Max” means “maximum” and, lastly, “ns” means “not significant”.

Feature	CG Mean	CG SD	CG Min	CG Max	KCG Mean	KCG SD	KCG Min	KCG Max	VAE-NTG Mean	VAE-NTG SD	VAE-NTG Min	VAE-NTG Max	Kruskal Wallis test	CG versus VAE-NTG	CG versus KCG	KCG versus VAE-NTG
F_0	7.77	0.235	7.27	8.5	7703	0.2257	7.27	8.27	6.86	0.5773	4.89	8.04	p<.0001	ns	p<.0001	p<.0001
F_1	484.5	74.97	302	807	387.3	76.62	259	566	196.8	75.06	0	512	p<.0001	p<.0001	p<.0001	p<.0001
F_2	878.1	174.4	543	1777	731.7	151.1	456	1249	396.4	188.9	0	1155	p<.0001	p<.0001	p<.0001	p<.0001
F_3	0.7131	0.506	-1.18	1.95	1472	0.6081	259	2.81	7193	3684	1.49	24.87	p<.0001	p<.0001	p<.0001	p<.0001
F_4	0.01888	0.6809	-3.39	1.43	0.765	0.6447	-0.84	1.91	2514	0.7846	-0.61	4.62	p<.0001	p<.0001	p<.0001	p<.0001
F_5	0.0319	0.685	-2.91	1.7	0.9195	0.7001	-0.71	2.09	2662	0.6863	-0.22	4.46	p<.0001	p<.0001	p<.0001	p<.0001
F_6	-0.1481	0.6973	-1.43	2.48	0.03625	0.7702	-1.43	1.66	6921	5003	-1.22	26.32	p<.0001	ns	p<.0001	p<.0001
F_7	0.3283	0.6729	-1.37	1.96	0.8577	0.7399	-1.5	2.27	8.4	4453	0.38	26.94	p<.0001	p=.0004	p<.0001	p<.0001
F_8	0.03883	0.8957	-1.72	3.23	0.4053	0.8073	-1.41	1.97	8114	6214	0.08	38.35	p<.0001	ns	p<.0001	p<.0001
F_9	-0.2878	0.8057	-2.48	2.64	0.5236	0.9665	-1.23	3.14	2466	1731	-0.83	10.67	p<.0001	p<.0001	p<.0001	p<.0001
F_{10}	-0.289	0.868	-2.99	1.87	-0.0175	0.9046	-2.17	1.77	5157	3787	-0.59	21.84	p<.0001	ns	p<.0001	p<.0001
F_{11}	0.05655	0.7327	-2.25	2.13	0.9039	0.8635	-1.02	2.77	6462	4609	-6.12	35.25	p<.0001	p<.0001	p<.0001	p<.0001
F_{12}	-1481	1245	-5.58	1.6	-0.9133	0.9744	-4.56	0.79	2241	2615	-5.49	15.13	p<.0001	p=.01	p<.0001	p<.0001
F_{13}	0.8073	0.9357	-1.71	3.59	1299	0.9726	-0.91	3.77	1182	1364	-2.21	10.52	p=.0001	p=.001	p=.0078	ns
F_{14}	3997	0.2113	3.34	4.65	3815	0.2217	3.27	4.26	3532	0.2735	2.77	4.21	p<.0001	p<.0001	p<.0001	p<.0001
F_{15}	11.67	0.616	9.76	13.59	11.17	0.6321	9.66	12.49	10.56	0.7142	8.63	12.33	p<.0001	p<.0001	p<.0001	p<.0001
F_{16}	24.97	1322	20.8	29	23.97	1302	20.9	26.9	23	1419	19	27.3	p<.0001	p<.0001	p<.0001	p=.0002
F_{17}	60.56	3405	48.9	71	58.65	2.95	51.1	65.2	56.61	3322	47.6	66.6	p<.0001	p=.0002	p<.0001	p=.0007
F_{18}	1016	0.00265	1008	1024	1019	0.003465	1012	1026	1042	0.01846	0.9928	1167	p<.0001	p<.0001	p<.0001	p<.0001
F_{19}	558.3	29.42	467	650	532	31.07	457	595	486.8	39.15	371	580	p<.0001	p<.0001	p<.0001	p<.0001
F_{20}	1064	0.00812	1034	1086	1074	0.009585	1051	1095	1144	0.05458	1007	1491	p<.0001	p<.0001	p<.0001	p<.0001
F_{21}	584.6	30.4	489	679	560.5	31.33	485	627	533.2	35.64	437	616	p<.0001	p<.0001	p<.0001	p<.0001
F_{22}	1145	0.01734	1094	1203	1162	0.01917	1121	1211	1267	0.08429	1029	1764	p<.0001	p<.0001	p<.0001	p<.0001
F_{23}	629.3	32.62	525	731	606.3	32.59	531	679	587.4	52.54	0	718	p<.0001	p<.0001	p<.0001	p=.011
F_{24}	1272	0.03299	1176	1392	1.3	0.03492	1224	1376	1416	0.1051	1061	2019	p<.0001	p<.0001	p<.0001	p<.0001
F_{25}	698.9	37.03	578	804	678.2	34.83	595	753	656.1	56.86	0	799	p<.0001	p=.0002	p<.0001	p=.0036
F_{26}	5861	1736	2	12	7594	1974	4	13	39.41	21.25	7	144	p<.0001	p<.0001	p<.0001	p<.0001
F_{27}	-1316	2725	-11	8	-1078	3726	-10	8	23.92	21.95	-16	121	p<.0001	ns	p<.0001	p<.0001
F_{28}	0.9732	2255	-6	8	3344	2983	-5	10	36.69	21.55	-1	142	p<.0001	p<.0001	p<.0001	p<.0001
F_{29}	1569	0.6492	0	4	1953	0.8248	1	5	15.76	10.65	2	65	p<.0001	ns	p<.0001	p<.0001
F_{30}	0.09489	1117	-4	3	0.2969	1293	-3	3	9944	9744	-14	51	p<.0001	ns	p<.0001	p<.0001
F_{31}	0.007299	0.8625	-2	3	0.125	0.9419	-2	2	13.42	9965	-3	59	p<.0001	ns	p<.0001	p<.0001
F_{32}	4757	3692	0	20.1	0.3906	1063	-2	2	28.64	21.28	0.1	125.4	p<.0001	p=.0078	p<.0001	p<.0001
F_{33}	0.008246	0.00449	0.001	0.025	7161	4865	0	20	0.09098	0.05757	0.01	0.31	p<.0001	p=.001	p<.0001	p<.0001
F_{34}	16.95	5662	7	37	0.01258	0.006563	0.003	0.034	68.51	33.14	18	192	p<.0001	ns	p<.0001	p<.0001
F_{35}	0.09783	0.0429	0.02	0.24	18.47	4.5	9	31	0.6825	0.3836	0.15	2.02	p<.0001	p=.001	p<.0001	p<.0001
F_{36}	1019	0.01776	0.96	1.07	0.1325	0.05149	0.03	0.31	1175	0.1061	0.96	1.58	p<.0001	p=.0005	p<.0001	p<.0001
F_{37}	549.5	29.17	459	641	-0.5742	0.2532	-1.22	0	467.7	42.35	318	568	p<.0001	p<.0001	p<.0001	p<.0001
F_{38}	-0.4462	0.2433	-1.17	0.21	1033	0.01866	0.99	1.08	-0.5435	0.3547	-2.97	0.34	p=.001	p=.001	p=.006	ns
F_{39}	-2215	2401	-8.2	5.7	6247	0.2107	5.75	6.76	-9391	5041	-34.3	5.5	p<.0001	p=.01	p<.0001	p<.0001
F_{40}	0.1856	0.03151	0.122	0.325	-3311	1877	-6.6	3.7	0.606	0.2934	0.17	2.09	p<.0001	p=.0008	p<.0001	p<.0001
F_{41}	0.356	0.08243	0.17	0.63	0.2143	0.04321	0.145	0.337	2.11	1.24	0.409	7171	p<.0001	p=.0011	p<.0001	p<.0001
F_{42}	0.3467	0.0915	0.175	0.684	0.4359	0.1268	0.232	0.876	1857	1134	0.326	6261	p<.0001	p=.0092	p<.0001	p<.0001
F_{43}	0.9132	0.1086	0.57	1.22	0.4224	0.1846	0.188	1553	1861	0.6816	0	6.12	p<.0001	p<.0001	p<.0001	p<.0001
F_{44}	1155	0.152	0.76	1.74	1039	0.1278	0.75	1.31	2761	1599	1.08	21.71	p<.0001	p<.0001	p<.0001	p<.0001
F_{45}	0.6449	0.1054	0.32	0.97	1388	0.2164	0.96	1.85	1.31	1688	-22.54	5.1	p<.0001	p=.0002	p<.0001	p<.0001
F_{46}	-0.0268	0.03903	-0.134	0.085	0.7359	0.1207	0.44	1.02	0.4109	0.2893	-0.469	1213	p<.0001	p<.0001	p<.0001	p<.0001
F_{47}	-0.02394	0.1503	-0.391	0.463	0.0237	0.04871	-0.08	0.144	-1575	1.14	-5357	1816	p<.0001	p=.0011	p<.0001	p<.0001
F_{48}	-0.04494	0.1506	-0.455	0.452	-0.1816	0.1661	-0.544	0.261	-1388	1046	-5071	1728	p<.0001	p=.0002	p<.0001	p<.0001
F_{49}	0.002903	0.01465	-0.038	0.057	-0.173	0.1579	-0.515	0.231	-0.04916	0.05818	-0.265	0.169	p<.0001	p=.0003	p<.0001	p<.0001
F_{50}	-0.002822	0.02786	-0.103	0.082	0.01111	0.03179	-0.067	0.108	0.1549	0.2368	-0.812	0.979	p<.0001	ns	p<.0001	p<.0001

among the three classes. Fig. 6 shows the results for each possibility, also listed in part in the third column of Table 3: only one feature, subsets with n features for ($2 \leq n \leq 51$) and, lastly, the complete vectors with 52 features. The best option for each case, i.e., that which attracts $P = (G_1, G_2)$ closer to the corner (1,0) of the paraconsistent plane, reveals important facts about the features.

3.3. Classification results

Notably, to perform the classifications, our generic SVM structure defined in Section 2 and Fig. 5 was adapted for each one of the 52 cases listed in Table 3 in such a way that there might be $X = 1, 2, \dots, 52$ passive element in its input layer. Contrary to the adaption, the hidden layer and the output layer contain always $Q = \lfloor \frac{777}{2} \rfloor = 388$ active nonlinear elements and just one linear active element, respectively. The value chosen for Q corresponds to 50% of our database, as commonly adopted in related works.

Table 3 shows the best features, i.e., those which place $P = (G_1, G_2)$ closer to the corner (1,0) of the paraconsistent plane, for each one of the 52 possibilities shown in Fig. 6 and considering the distinction of our three classes, i.e., CG, KC and VAE-NTG, using a one-against-all approach. In addition, the corresponding best accuracies, sensibilities and specificities, obtained considering a hold-out procedure repeated 2000 times with Q testing cases, are also shown in Table 3.

Particularly interesting is the situation where only one feature, i.e., F_3 or BAD-D [35], was used to feed the SVM classifier, implying in values of accuracy, sensitivity and specificity of 0.813, 0.671 and 1.0, respectively. Another curious aspect is the fact that, as we increase the number of features, the results in the last three columns of Table 3 do not necessarily get better, implying that specific features are disadvantageous when considered in conjunction with some others.

In addition to the above-mentioned remarks, Tables 4 and 5 compare the results obtained considering just the classes CG and VAE-NTG in different scenarios: (i) CTMVI using all the 52 features, (ii) CTMVI

Table 3

Best features for each option (only 1 feature, 2 features, ..., 52 features) and the corresponding distances from $P = (G_1, G_2)$ to the corner (1,0) of the paraconsistent plane, in addition to accuracies, sensibilities and specificities. Our three classes were considered, i.e., CG, KC and VAE-NTG, based on the one-against-all procedure.

Number of features	Features	Distance from P to (1,0)	Accuracy	Sensitivity	Specificity
1	F_3	1.597	0.813	0.671	1.000
2	F_3, F_{28}	1.649	0.821	0.694	1.000
3	F_3, F_{26}, F_{28}	1.644	0.810	0.695	1.000
4	$F_3, F_{26}, F_{28}, F_{29}$	1.677	0.808	0.675	1.000
5	$F_3, F_{26}, F_{27}, F_{28}, F_{29}$	1.705	0.805	0.677	1.000
6	$F_3, F_{26}, F_{27}, F_{28}, F_{29}, F_{31}$	1.726	0.808	0.675	1.000
7	$F_3, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}$	1.740	0.797	0.642	1.000
8	$F_3, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{44}$	1.743	0.801	0.648	1.000
9	$F_3, F_{10}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{44}$	1.745	0.803	0.642	1.000
10	$F_3, F_5, F_{10}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{44}$	1.740	0.801	0.669	1.000
11	$F_3, F_4, F_5, F_{10}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{44}$	1.745	0.813	0.679	1.000
12	$F_3, F_4, F_5, F_7, F_{10}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{44}$	1.752	0.821	0.694	1.000
13	$F_3, F_4, F_5, F_7, F_{10}, F_{12}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{44}$	1.762	0.816	0.681	1.000
14	$F_3, F_4, F_5, F_7, F_{10}, F_{12}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{40}, F_{44}$	1.771	0.818	0.683	1.000
15	$F_3, F_4, F_5, F_6, F_7, F_{10}, F_{12}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{40}, F_{44}$	1.768	0.816	0.686	1.000
16	$F_3, F_4, F_5, F_6, F_7, F_{10}, F_{11}, F_{12}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{40}, F_{44}$	1.772	0.816	0.686	1.000
17	$F_3, F_4, F_5, F_6, F_7, F_8, F_{10}, F_{11}, F_{12}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{40}, F_{44}$	1.776	0.813	0.675	1.000
18	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{40}, F_{44}$	1.780	0.813	0.684	1.000
19	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{40}, F_{44}, F_{49}$	1.783	0.811	0.677	1.000
20	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{40}, F_{44}, F_{47}, F_{49}$	1.781	0.831	0.685	1.000
21	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{40}, F_{41}, F_{44}, F_{47}, F_{49}$	1.775	0.819	0.696	1.000
22	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{40}, F_{41}, F_{43}, F_{44}, F_{47}, F_{49}$	1.772	0.816	0.686	1.000
23	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{47}, F_{49}$	1.775	0.819	0.683	1.000
24	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{47}, F_{48}, F_{49}$	1.777	0.816	0.681	1.000
25	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{47}, F_{48}, F_{49}, F_{51}$	1.780	0.813	0.671	1.000
26	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.784	0.818	0.663	1.000
27	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{14}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.788	0.813	0.675	1.000
28	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{14}, F_{15}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.792	0.811	0.665	1.000
29	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.798	0.806	0.669	1.000
30	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.793	0.813	0.675	1.000
31	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{20}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.794	0.813	0.667	1.000
32	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{20}, F_{22}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.796	0.816	0.665	1.000
33	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{20}, F_{22}, F_{24}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.799	0.811	0.673	1.000
34	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{20}, F_{22}, F_{24}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{33}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.793	0.805	0.660	1.000
35	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{20}, F_{22}, F_{24}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{33}, F_{37}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.794	0.801	0.656	1.000
36	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{19}, F_{20}, F_{22}, F_{24}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{33}, F_{37}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.797	0.808	0.667	1.000
37	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{19}, F_{20}, F_{22}, F_{24}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{33}, F_{35}, F_{37}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.793	0.808	0.662	1.000
38	$F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{19}, F_{20}, F_{22}, F_{24}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{33}, F_{34}, F_{35}, F_{37}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.794	0.808	0.671	1.000
39	$F_0, F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{19}, F_{20}, F_{22}, F_{24}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{33}, F_{34}, F_{35}, F_{37}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.793	0.808	0.675	1.000
40	$F_0, F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{19}, F_{20}, F_{22}, F_{24}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{33}, F_{34}, F_{35}, F_{37}, F_{39}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.793	0.808	0.671	1.000

(continued on next page)

Table 3 (continued).

Number of features	Features	Distance from P to $(1,0)$	Accuracy	Sensitivity	Specificity
41	$F_0, F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{19}, F_{20}, F_{22}, F_{24}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{33}, F_{34}, F_{35}, F_{36}, F_{37}, F_{39}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.794	0.813	0.684	1.000
42	$F_0, F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{19}, F_{20}, F_{22}, F_{24}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{33}, F_{34}, F_{35}, F_{36}, F_{37}, F_{39}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{45}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.795	0.813	0.679	1.000
43	$F_0, F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{19}, F_{20}, F_{22}, F_{24}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{33}, F_{34}, F_{35}, F_{36}, F_{37}, F_{39}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{45}, F_{46}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.794	0.842	0.717	1.000
44	$F_0, F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{19}, F_{20}, F_{22}, F_{24}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{32}, F_{33}, F_{34}, F_{35}, F_{36}, F_{37}, F_{39}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{45}, F_{46}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.796	0.852	0.732	1.000
45	$F_0, F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{19}, F_{20}, F_{21}, F_{22}, F_{24}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{32}, F_{33}, F_{34}, F_{35}, F_{36}, F_{37}, F_{39}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{45}, F_{46}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.799	0.842	0.721	1.000
46	$F_0, F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{19}, F_{20}, F_{21}, F_{22}, F_{23}, F_{24}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{32}, F_{33}, F_{34}, F_{35}, F_{36}, F_{37}, F_{39}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{45}, F_{46}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.802	0.855	0.734	1.000
47	$F_0, F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{19}, F_{20}, F_{21}, F_{22}, F_{23}, F_{24}, F_{25}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{32}, F_{33}, F_{34}, F_{35}, F_{36}, F_{37}, F_{39}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{45}, F_{46}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.805	0.852	0.750	1.000
48	$F_0, F_1, F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{19}, F_{20}, F_{21}, F_{22}, F_{23}, F_{24}, F_{25}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{32}, F_{33}, F_{34}, F_{35}, F_{36}, F_{37}, F_{39}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{45}, F_{46}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.805	0.855	0.738	1.000
49	$F_0, F_1, F_2, F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{19}, F_{20}, F_{21}, F_{22}, F_{23}, F_{24}, F_{25}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{32}, F_{33}, F_{34}, F_{35}, F_{36}, F_{37}, F_{39}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{45}, F_{46}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.805	0.858	0.767	1.000
50	$F_0, F_1, F_2, F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{18}, F_{19}, F_{20}, F_{21}, F_{22}, F_{23}, F_{24}, F_{25}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{32}, F_{33}, F_{34}, F_{35}, F_{36}, F_{37}, F_{38}, F_{39}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{45}, F_{46}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.809	0.860	0.759	1.000
51	$F_0, F_1, F_2, F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{16}, F_{18}, F_{19}, F_{20}, F_{21}, F_{22}, F_{23}, F_{24}, F_{25}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{32}, F_{33}, F_{34}, F_{35}, F_{36}, F_{37}, F_{38}, F_{39}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{45}, F_{46}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.812	0.855	0.752	1.000
52	$F_0, F_1, F_2, F_3, F_4, F_5, F_6, F_7, F_8, F_9, F_{10}, F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{16}, F_{17}, F_{18}, F_{19}, F_{20}, F_{21}, F_{22}, F_{23}, F_{24}, F_{25}, F_{26}, F_{27}, F_{28}, F_{29}, F_{30}, F_{31}, F_{32}, F_{33}, F_{34}, F_{35}, F_{36}, F_{37}, F_{38}, F_{39}, F_{40}, F_{41}, F_{42}, F_{43}, F_{44}, F_{45}, F_{46}, F_{47}, F_{48}, F_{49}, F_{50}, F_{51}$	1.815	0.862	0.751	1.000

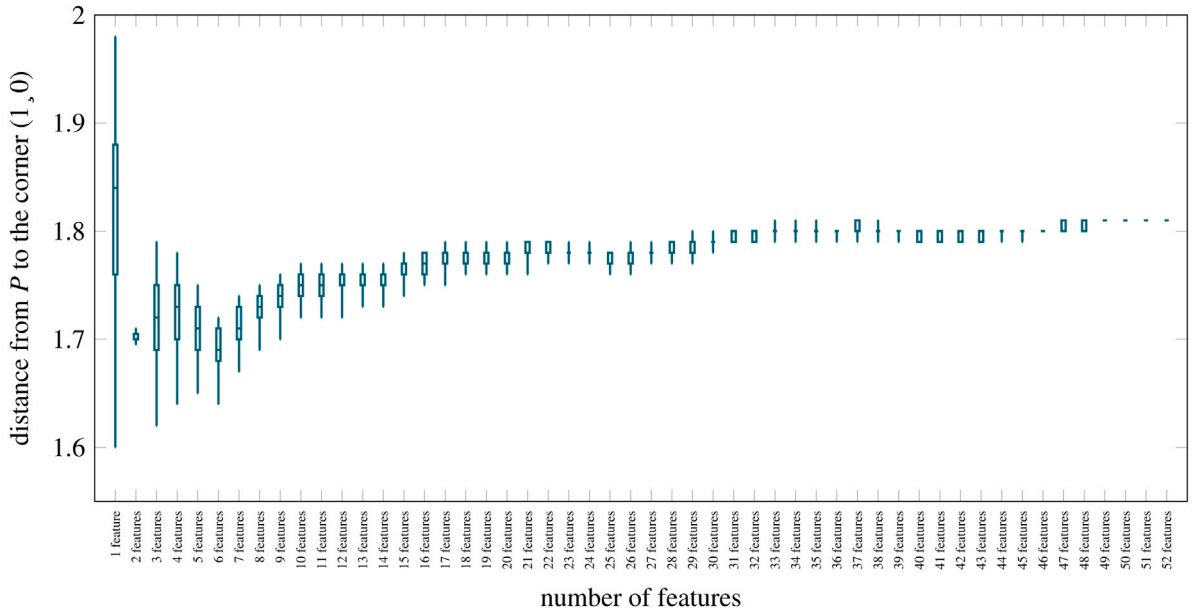
Fig. 6. Box-plots derived from PFE-based analyses used to find the best S -sample long subsets of features, where $(1 \leq S \leq 52)$.

Table 4

Sensitivity, specificity and AUC obtained with CTMVI using BAD-D as the sole feature, CTMVI using all the 52 features and PRFI, in isolation, distinguishing patients with CG from those with VAE-NTG.

	Sensitivity	Specificity	AUC
CTMVI with BAD-D only	0.594	0.924	0.824
CTMVI with 52 features	0.844	0.874	0.926
PRFI	0.718	0.905	0.862

Table 5

Sensitivity and specificity obtained with CTMVI using all the 52 features, CTMVI using BAD-D as the sole feature and PRFI, combined in sequential and parallel modes, distinguishing patients with CG from those with VAE-NTG.

	Sensitivity in parallel test	Specificity in parallel test	Sensitivity in serial test	Specificity in serial test
CTMVI + PRFI	0.99	0.84	0.80	0.99
CTMVI + BAD-D	0.98	0.86	0.75	0.99
BAD-D + PRFI	0.98	0.84	0.75	0.99

using just BAD-D as the sole feature, (iii) PRFI and (iv) the pairwise combinations of the previous possibilities adopting parallel and sequential procedures [20], where an increase in sensitivity related to the parallel approach forces specificity to drop and vice-versa. Observably, the pairwise combinations are those which provide the best results. Complementarily, Figs. 7–9 show pairwise comparisons involving the values of CTMVI using all the 52 features, CTMVI using just BAD-D as the sole feature and PRFI, considering the distinction between CG and VAE-NTG. The CTMVI of CG vs. VAE-NTG was then compared with the BAD D of CG vs. VAE-NTG, with the AUC of CTMVI being significantly higher, i.e., 0.935 (87.50% sensitivity (se), 84.95% specificity (sp)), with a CTMVI cutoff value of ≤ 0.639 , than that of BAD D, i.e., 0.765 (78.12% se, 86.05% sp), with a BAD D cutoff value of > 0.9 . The difference in AUROC between CTMVI and BAD D was 0.170, and the difference was significant ($p = 0.0001$). The CTMVI of CG vs. VAE-NTG was compared with the PRFI of CG vs. VAE-NTG, with the AUC of CTMVI being significantly higher, i.e., 0.935, than that of PRFI, i.e., 0.805, (59.38% se, 90.78% sp), with a PRFI cutoff value of > 0.12 . The difference in AUC between CTMVI and PRFI was significant at 0.130 ($p = 0.0004$). The difference in the AUCs of PRFI and BAD D was not significant. The comparison between the CTMVI, PRFI, and BAD D of CG vs. KCG showed no significant difference. The CTMVI, BAD D, and PRFI of CG and KCG showed an AUC of 1.0 (100% se, 100% sp), with a cutoff value of ≤ -1.36 ; AUC of 1.0 (100% se, 98.54% sp), with a cutoff value > 1.66 ; and AUC of 1.0 (100% se, 100% sp), with a cutoff value of > 0.374 , respectively.

Notably, the CTMVI was applied to the 237 patients from the external validation between individuals with healthy corneas CG ($n = 205$) and VAE-NTG ($n = 32$). The test showed 0.918 AUC with 84.37% se, 88.29% sp, 87.76% accuracy, 52.9% positive predictive value (PPV), and 97.31% negative predictive value (NPV) with a cutoff value of ≤ 0.57 compared with CG and VAE-NTG. The BAD D for these same patients showed 0.884 AUC with 71.87% se 93.66% sp, 90.71% accuracy, 63.86% PPV, and 95.52% NPV, with a cutoff value of > 1.4 compared with CG and VAE-NTG. The PRFI for these same patients showed 0.923 AUC with 84.37% se, 93.65% sp, 92.40% accuracy, 60.00% PPV and 97.39% NPV, with a cutoff value of > 0.118 compared with CG and VAE-NTG. Fisher's exact test did not show significant differences between CTMVI, BAD D, and PRFI, thus this reassures the efficacy of CTMVI.

4. Discussion

Progressive thinning and ectasia, i.e., worrisome ocular issues, might be exacerbated as a function of corneal refractive surgery and

eye rubbing [36], with genetic inheritance seemingly expressed in a variable manner [36,37]. Thus, preoperative screening is of paramount importance for refractive surgery candidates in order to exclude conditions that predispose to iatrogenic ectasia [32]. The principal independent risk factor is preexisting corneal changes similar to ectasia, i.e., pellucid marginal corneal degeneration or KC [7,15,38,39] or KC. Another possible problem is the lack of consensus on how to define those cases, which often hinders the comparison of results from different studies. Several acronyms have been used in order to describe corneas with early manifestations of this condition, including those with forme fruste keratoconus (FFKC), preclinical or subclinical keratoconus and keratoconus suspect (KCS). In order to avoid overlapping definitions of FFKC and KCS, we decided to adopt the term “VAE-NT” to refer to the eye with normal topography in very asymmetric cases, as presented by other authors [21,31]. Considering that suspected cases are diagnosed more easily because they already show topographical signs of the disease, the essential clinical challenge is detecting susceptible cases that are not identifiable by current screening methods but are predisposed to ectasia and with possible progression to the clinically evident condition [40].

According to our experiments, the 3rd order vertical coma of the back surface, i.e., F_{47} or Z 3 -1 CB, showed better sensitivity and specificity to distinguish both HSCE and KC cases from normal cases among the variables assessing these aberrations, in contrast to the results of the study conducted by Hashemi et al. [41]. Being slightly different from our findings, those authors mentioned that the 5th order vertical aberration of the front surface, i.e., F_{50} or Z 5 -1 CF, had the best diagnostic validity to detect subclinical keratoconus considering all Pentacam aberration indices. This is what can be observed in Table 3 when F_{47} appears as being a contributing parameter when the number of features is at least 20, in contrast to F_{50} appearing when the number of features is at least 26, i.e., the former is preferable than the latter. Numerous studies have proposed the use of higher-order aberrations to diagnose and classify keratoconus [42]. Some recommend that the aberration and vertical asymmetry topometric indices should be assessed simultaneously for subclinical KC [41,43]. In addition, Gordon-Shaag [43] suggests that although these indices cannot be used for refractive surgery screening, they should be assessed routinely in addition to the thickness and curvature of the cornea to detect subclinical keratoconus, especially when the chances of a post-LASIK ectasia are relatively high.

Previous studies have used AI for the same purpose [21,44–47]. Two of them involved the use of different AI methods to identify VAE-NT by using a Pentacam HR-based strategy. Particularly, Ruiz Hidalgo et al. [44] used an SVM-derived model and reported an AUC slightly lower than 0.922, with 79.1% sensitivity and 97.9% specificity. Kovács et al. [45] used bilateral data and artificial neural networks from patients. They obtained an AUC of 0.96, with 92% sensitivity and 85% specificity, a result similar to that obtained by Lopes et al. [21], where the AUC is registered as 0.968 by using PRFI. However, the former study included no independent test, which the authors considered a negative point. Arbelaez et al. [46] used an SVM model to diagnose subclinical keratoconus and achieved 92% sensitivity and 97.7% specificity. Indeed, their results cannot be straightforwardly compared with those we obtained because their definition of keratoconus differs from our definition of SKC (VAE-NT). Bae et al. [47], using Pentacam HR unique parameters, obtained the best accuracy in VAE-NT detection with the superotemporal versus inferonasal asymmetry index (ST-IN), reflecting an AUC of 0.809. Other studies applied cross validation test. Lopes et al. [21] showed that PRFI had an AUC of 0.992, i.e., 0.942 se, 0.988 sp and 0.216 cutoff, being statistically higher than the BAD-D index; AUC of 0.96, i.e., 0.873 se, 0.975 sp and $p = 0.006$, when compared the normal eyes versus KC eyes with VAE-NT eyes included. The detection of normal eyes versus VAE-NT eyes alone presented 0.85 se and 0.96 sp considering only the independent test set with optimized cutoff of 0.125. This study used the holdout cross validation

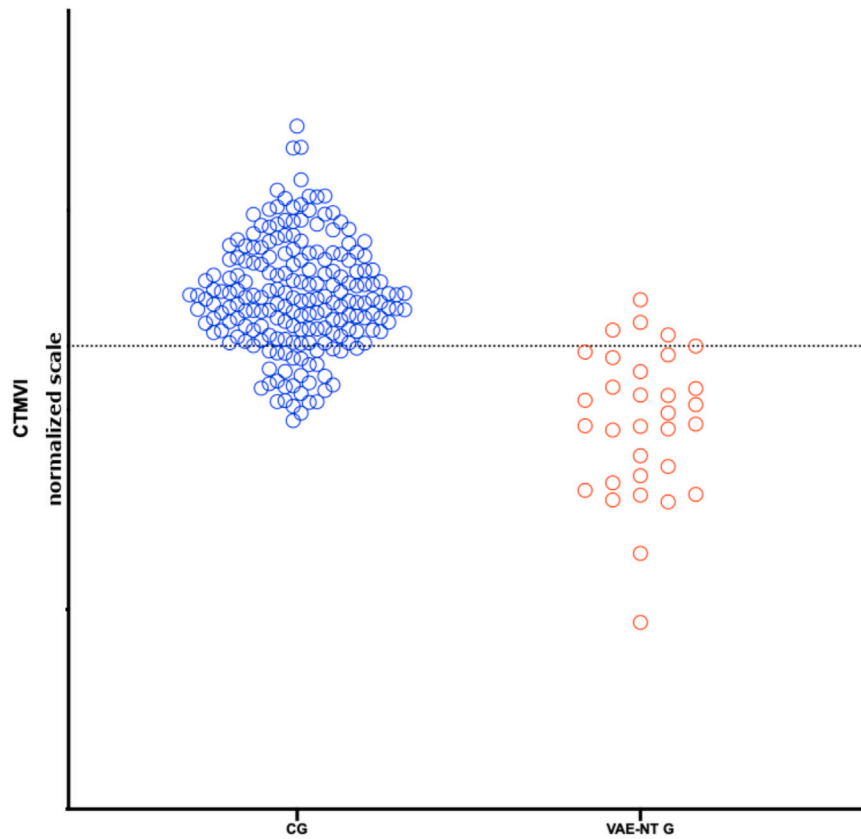


Fig. 7. Pairwise comparison between CG and VAE-NTG for CTMVI considering all the 52 features.

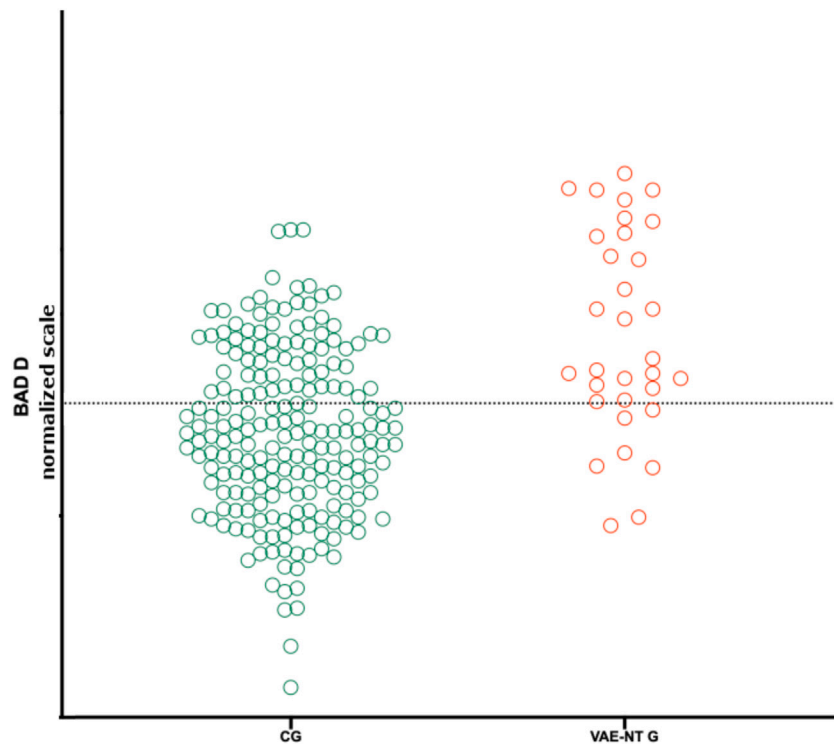


Fig. 8. Pairwise comparison between CG and VAE-NTG for CTMVI considering just BAD-D as the sole feature.

method confronted $\frac{2}{3}$ of cases in the training data set and $\frac{1}{3}$ of cases in external validation set [21]. However, in this situation the VAE-NT cases were all from independent test, implying they were not

used in the training set [21]. Our study obtained 0.87 se, 0.90 sp and 0.89 accuracy (acc), but in contrast we used tomography criteria to increase the control of the enrollment of patients into CG and

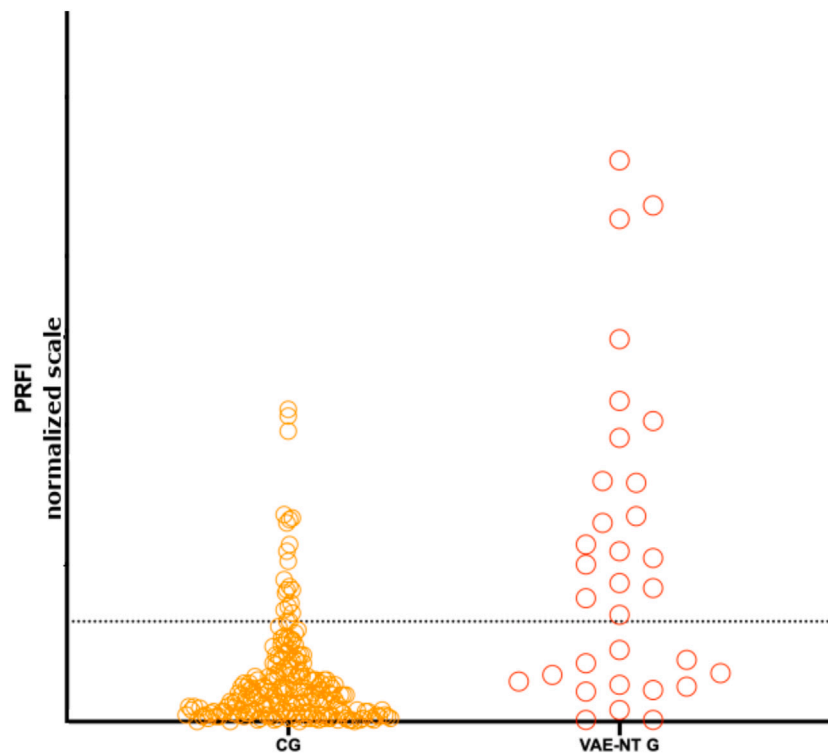


Fig. 9. Pairwise comparison between CG and VAE-NTG considering PRFI strategy.

VAE-NTG. Ambrósio et al. [32] adopted the validation model of leave-one-out cross-validation (LOOCV) to enhance ectasia detection through Tomography and Biomechanical Index (TBI), revealing 0.90 se and 0.96 sp, comparing NG (normal group) and VAE-NTG, although they used only topography criteria to categorize the patients of these two groups. Saad and Gatinel [48] described a linear discriminant-based model that was developed with another corneal tomography (Orbscan II; Technolas, Munich, Germany), with 0.93 se and 0.92 sp. This model was further validated in two independent populations. In one study published by the authors the sensitivity/specificity were similar to the original, i.e., 0.92 and 0.96 respectively [49]. In the other, the sensitivity/specificity were 0.70 and 0.96 respectively, although a different ethnic group was evaluated [50]. AI methods for data analysis has gained continuous momentum in the last decade and, thus, promising results have been described. Consequently, it is of fundamental importance to learn how to interpret and use all these data in order to make better clinical decisions for the patient benefit [51].

Overall, CTMVI was shown to have good sensitivity and specificity, in line with parameters that have been used in clinical practice and also described in the literature [20,21,52,53]. Particularly, since sensitivity is much more important in such a clinical diagnoses, the parallel approach involving CTMVI with 52 features and PRFI, listed in the first row of Table 5, can be considered the preferable strategy with an excellent potential. In the internal validation the CTMVI of CG was 1.11 ± 0.46 and showed significant difference when compared with the value of -1.04 ± 1.41 VAE-NTG, and KCG -16.24 ± 10.02 ($p \leq .0001$). BAD D and PRFI showed similar results.

The main limitations of our study consist of its small sample size and, in addition, the lack of long-term follow-up. Additionally, VAE-NTG could be studied longitudinally to confirm whether the condition will definitively progress to a KC or it is a normal cornea of unilateral keratoconus. Our sample size was considerably limited by the inclusion criteria adopted, but we believe it was better to assess a truly sub-clinical population whose less affected eyes were, indeed, not readily identifiable in a subjective manner or by any technology in isolation. This is because more advanced degrees of KC and corneal ectasia are already identified by a number of technologies.

Another limitation of this study was that corneal biomechanical parameters as Ocular Response Analyzer or Corvis were not used. The Tomography and Biomechanical Index (TBI) derived from Corvis and Pentacam provided sensitivity/specificity for screening topographical normal subclinical keratoconic eyes from healthy eyes of 0.90/0.96 but in this study only topographic criteria were used to enroll patients for VAE-NT group [32]. Other study used topographic and tomographic criteria for VAE-NT group ($BAD < 1.69$) similar to our study, but in this case the TBI showed 0.71 se, 0.72 sp and 0.71 acc [53]. An additional limitation consists of the fact that our CG included patients who have undergone LASIK and those who have undergone PRK. Ectasia may take longer to manifest after PRK, as reported in the literature for situations where it occurred 9 years after surgery [54]. Post-PRK ectasia is extremely rare [55] compared with post-LASIK ectasia [56]. Although this is expected to be relatively rare, it is possible that some eyes with a normal clinical examination (including corneal topography and tomography) have mild or susceptible forms of ectasia, there were cases that progressed to corneal ectasia after different laser vision correction (LVC) procedures [57–60]. In this study we used rigorous criteria to reinforce the classification of normal cornea values, including not only topography standards, but also tomography. The preoperative with long-term follow-up after laser LVC would provide a more robust population for the normal group [48,61,62].

In a retrospective case series, Bohac et al. [63] analyzed data from 30167 patients and found ectasia in ten eyes of seven patients (0.033%). These data corroborate the idea that the incidence of ectasia dropped 0.66%, which was reported by the work of Pallikaris in 2001 [64]. This is most likely the result of important developments related to advanced and notable screening strategies. However, mysterious cases of ectasia are still a challenge, stimulating further research in the field. In fact, eye allergies with consequent rubbing may be a triggering factor for ectasia [16], most probably in patients genetically predisposed to KC [65]. Nevertheless, studies have also indicated that eye rubbing may play a decisive role in the development of corneal ectasia in keratoconus [65,66], but KC may be caused by self-induced or accidental ocular injuries [67]. The widely accepted risk factors are

not rigorous enough to determine the potential development of post-LASIK ectasia. Thus, there is a hope and a need to increase accuracy, with better specificity and sensitivity for these factors [63]. Ectasia susceptibility indeed depends on resemblance to VAE-NT, corneal instability and surgical parameters such as flap thickness/ablation depth, requiring a better understanding [10,11,13,68].

With the constant development and advancements in diagnoses, there has been an unmet demand for noninvasive methods to screen and diagnose KC, especially HSCE. Given that its etiology is not fully known yet, new screening methods are essential to improve early diagnosis, investigate treatment results and stimulating the development of new treatments. The good sensitivity and specificity of CTMVI in the discrimination between pathologically-affected patients and healthy controls obtained in our study indicate that it can be useful as a screening test to detect HSCE prior to laser refractive surgery based on LASIK, PRK or small-incision lenticule extraction. A small proportion of patients who undergo this type of surgery, unfortunately, develop corneal ectasia, even with preoperative tomography measurements that are considered normal.

In future studies, CTMVI may be used in combination with different indexes and diverse technologies to improve important works such as that performed by Hwang et al. [69], who evaluated the combination of Scheimpflug and spectral domain OCT analysis to assess corneas with asymmetrical keratoconus. The CTMVI algorithm does not measure corneal biomechanics directly but describes its morphology. It will remain difficult to identify the first manifestations of corneal ectasia until corneal biomechanics can be directly and precisely assessed. Recent studies including this assessment are promising but are still being robustly validated [70] or developed for clinical use [71]. A way to improve CTMVI efficacy is to increase the sample set, always considering sample independency and that the samples come from the same distribution. In this case, two trends are expected: one is that the distribution of the set, i.e., observed or empirical distribution, becomes closer to the real distribution and the other is that the model error decreases with gradual CMTVI improvements because our database will be updated with every new sample in the set. The sample set should not be too large to avoid SVM overfitting because SVM complexity also increases with sample size [25,26], as is explained by the law of large numbers from probability theory. Another strategy to improve CMTVI is the integration with other direct and indirect biomechanical parameters [53,69,71].

5. Conclusions

In conclusion, we described a PFE- and SVM-based AI model using tomography data to distinguish between healthy, KC and VAE-NT cases. CTMVI, mainly in parallel strategy associated with PRFI, showed excellent accuracy compared with other strategies previously reported in the literature. As a future work, we intend to improve the accuracy of the proposed approach for the detection of subclinical keratoconus and susceptible cases by increasing the dataset and assessing direct and indirect biomechanical features.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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